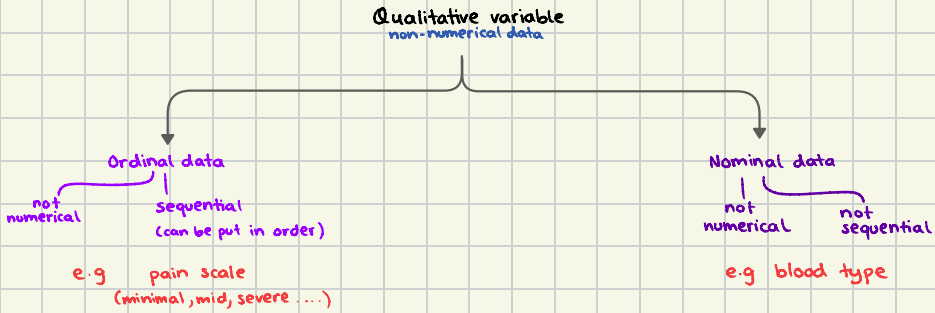
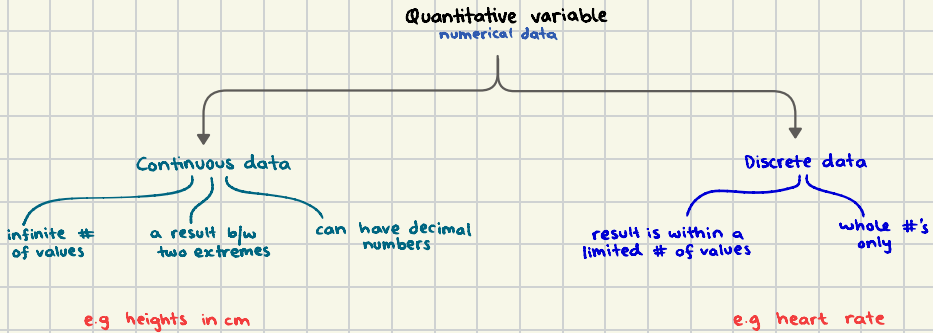
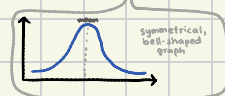
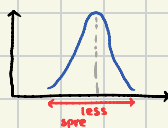
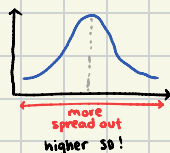


Identifying different types of variables



Standard deviation : tells us how a data is spread out from its mean value & it involves data with normal distribution

e.g



ad out
Lower SD!

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

standard deviation

Sum of

individual set values

mean

total # of observations

Petal lengths / cm			
3.2	2.7	3.1	3.0
3.1	3.3	3.3	3.2
2.9	3.4	2.9	3.0

- ① List observations (x)
- ② calculate the mean ($\bar{x}$)
- ③ calculate difference b/w each observation & mean
- ④ calculate the square of each difference
- ⑤ calculate sum of all the squares of each difference
- ⑥ calculate standard deviation

① x	③ (x - $\bar{x}$)	④ (x - $\bar{x}$) ²
3.2	0.11	0.0121
2.7	-0.39	0.1521
3.1	0.01	0.0001
3.0	-0.09	0.0081
3.1	0.01	0.0001
3.3	0.21	0.0441
3.3	0.21	0.0441
3.2	0.11	0.0121
2.9	-0.19	0.0361
3.4	0.31	0.0961
2.9	-0.19	0.0361
3.0	-0.09	0.0081

$$\sum x = 37.1$$

$$n = 12$$

$$\bar{x} = \frac{37.1}{12} = 3.09$$

$$\sum (x - \bar{x})^2 = 0.43462$$

$$n-1 = 11$$

$$\frac{\sum (x - \bar{x})^2}{n-1} = 0.0395$$

$$s = \sqrt{0.0395} = 0.199$$

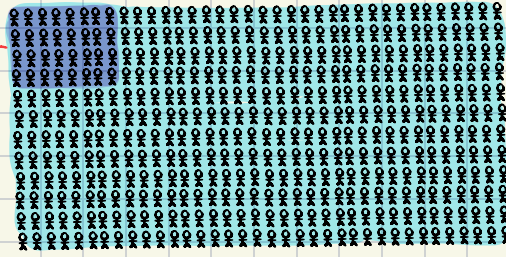
Petal length is 3.09 ± 0.199 cm

Standard error (SE): calculates how close the sample mean to the true mean (entire population's mean)

easy-to-get
but not completely
representative of
the entire population

sample
mean

sample
population
(50 people)



entire
population → true mean

e.g Sample mean = 6.5 ± 0.9

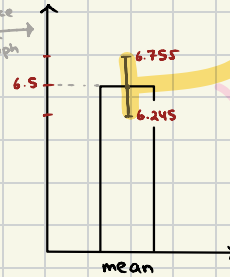
$$\text{standard error} = \frac{s}{\sqrt{n}} \rightarrow \begin{matrix} \text{sd} \\ \text{sample size} \end{matrix}$$

$$\text{standard error} = \frac{0.9}{\sqrt{50}} = \underline{\underline{0.127}}$$

≠ calculate $2 \times \text{SE}$

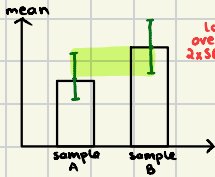
$$\hookrightarrow 0.127 \times 2 = \underline{\underline{0.255}}$$

make
a
bar
graph



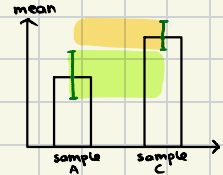
we are 95% certain
that the mean of the
entire population will be
within this range

smaller the range,
more reliable



larger
overlap in
 $2 \times \text{SE}$ bars = means are not
significantly
different

no
overlap in
 $2 \times \text{SE}$ bars = means are
significantly
different



best way to compare mean? No, best is t-test

T-test: statistical test to compare the means of two sets of data to see if they are significantly different

the two sets of data should be continuous, quantitative, showing normal distribution & >10 data should be present for each set

e.g	plant height using Fertiliser A /cm	plant height using Fertiliser B /cm
mean, $\bar{x}$	5.9	7.5
standard deviation, s	0.71	0.52
# of plants measured (data)	11	11

① construct a null hypothesis

→ there is No significant difference in the mean height of the plants in the two fields

② calculate the t-value

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$\bar{x}_1$ = mean of sample 1

$\bar{x}_2$ = mean of sample 2

s_1 = sd of sample 1

s_2 = sd of sample 2

n_1 = # of individual measurements in sample 1

n_2 = # of individual measurements in sample 2

$$t = \frac{|5.9 - 7.5|}{\sqrt{\frac{(0.71)^2}{11} + \frac{(0.52)^2}{11}}} = \underline{\underline{6.03}} \leftarrow \text{t-value}$$

③ calculate the degree of freedom

$$D.O.F = (n_1 - 1) + (n_2 - 1)$$

$$= (11 - 1) + (11 - 1)$$

WTF is this??
= 20

④ find critical value at $p=0.05$

$$t = 2.09$$

less certain	more certain	more certain	more certain
$p=0.10$	$p=0.05$	$p=0.025$	$p=0.01$

if t-value calculated is HIGHER than the value in the table	if t-value calculated is LOWER than the value in the table
→ the two data sets are significantly diff.	→ the two data sets are not significantly diff.
→ Reject null hypothesis	→ Accept null hypothesis
difference is not due to random error/chance	difference is due to random error/chance

calculated t-value table t-value

$$6.03 > 2.09$$

greater

∴ REJECT null hypothesis

Degrees of Freedom	p=0.05	p=0.025	p=0.01	p=0.005
1	12.71	25.45	63.66	127.32
2	4.30	6.20	9.92	14.09
3	3.18	4.17	5.84	7.45
4	2.78	3.50	4.60	5.60
5	2.57	3.16	4.03	4.77
6	2.45	2.97	3.71	4.32
7	2.36	2.84	3.50	4.03
8	2.31	2.75	3.36	3.83
9	2.26	2.68	3.25	3.69
10	2.23	2.63	3.17	3.58
11	2.20	2.59	3.11	3.50
12	2.18	2.56	3.05	3.43
13	2.16	2.53	3.01	3.37
14	2.14	2.51	2.98	3.33
15	2.13	2.49	2.95	3.29
16	2.12	2.47	2.92	3.25
17	2.11	2.46	2.90	3.22
18	2.10	2.44	2.88	3.20
19	2.09	2.43	2.86	3.17
20	2.09	2.42	2.84	3.15

Chi-squared (χ^2) test: statistical test to measure if the observed & expected values are significantly different
 the data should be quantitative & discrete (not continuous)

e.g. flip a coin 20 times

expected	observed
10 Tails	8 Tails
10 Heads	12 Heads

$$\chi^2 = \sum \frac{\overset{\text{observed}}{O} - \overset{\text{expected}}{E}}{E}^2$$

what is it used for? breeding experiments & ecological sampling



phenotype	expected	observed
	25	27
	25	22
	25	26
	25	25

① construct a null hypothesis

observed & expected results are NOT significantly different

② calculate the degree of freedom

$$\begin{aligned} \text{D.O.F} &= n - 1 \\ &= 4 - 1 \\ &= \underline{\underline{3}} \end{aligned}$$

n = possible phenotypes

③ calculate the χ^2 value

$$\begin{aligned} \chi^2 &= \sum \frac{(O-E)^2}{E} \\ \chi^2 &= \frac{(27-25)^2}{25} + \frac{(22-25)^2}{25} + \frac{(26-25)^2}{25} + \frac{(25-25)^2}{25} \\ \chi^2 &= 0.56 \end{aligned}$$

④ find critical value at $p=0.05$

$$\chi^2 = 7.81$$

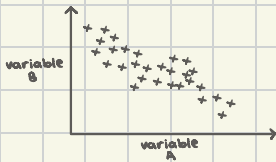
if χ^2 value calculated is HIGHER than the value in the table → expected & observed are significantly diff. → Reject null hypothesis difference is not due to random error/chance	if χ^2 value calculated is LOWER than the value in the table → expected & observed are not significantly diff. → Accept null hypothesis difference is due to random error/chance
---	--

calculated χ^2 value: 0.56
 table χ^2 value: 7.81
 0.56 < 7.81
 lesser

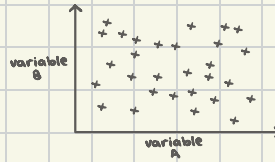
∴ ACCEPT null hypothesis

DF	.1	.05	.025	.01
1	2.71	3.84	5.02	6.63
2	4.61	5.99	7.38	9.21
3	6.25	7.81	9.35	11.34
4	7.78	9.49	11.14	13.28
5	9.24	11.07	12.83	15.09
6	10.64	12.59	14.45	16.81
7	12.02	14.07	16.01	18.48
8	13.36	15.51	17.53	20.09
9	14.68	16.92	19.02	21.67
10	15.99	18.31	20.48	23.21
11	17.28	19.68	21.92	24.72
12	18.55	21.03	23.34	26.22
13	19.81	22.36	24.74	27.69
14	21.06	23.68	26.12	29.14
15	22.31	25.00	27.49	30.58

Correlation test: to measure if there is a relationship b/w two variables



Negative correlation

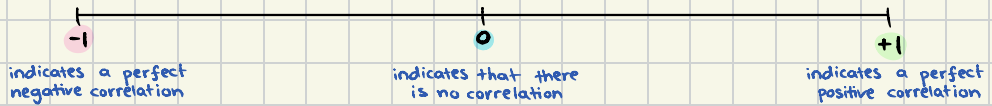


No correlation



Positive correlation

Strength of the correlation can be assessed by a correlation coefficient



Pearson's linear correlation coefficient :

the data should be continuous, normally distributed, scatter graph shows a linear correlation & atleast 5 paired data

x is variable X
 y is variable Y

$\bar{x}$ is mean of variable X
 $\bar{y}$ is mean of variable Y

$$r = \frac{\sum xy - n\bar{x}\bar{y}}{(n-1)s_x s_y}$$

n = # of readings

s_x = sd of x
 s_y = sd of y

(X)	(Y)	
# of plant A	# of ladybugs	x y
20	32	640
18	28	504
24	40	960
16	25	400
17	23	391
30	60	1800
26	48	1248

① mean, $\bar{x} = 21.57$

② $\bar{y} = 36.57$

③ $\sum xy = 5943$

$s_x = 5.22$

$s_y = 13.56$

① construct a null hypothesis

there is no significant correlation b/w the variables

② calculate the r value

$$r = \frac{\sum xy - n\bar{x}\bar{y}}{(n-1)s_x s_y}$$

$$r = \frac{5943 - 5521.7}{(7-1)(5.22)(13.56)}$$

r = 0.992

strong positive correlation

③ calculate the degree of freedom

$$D.O.F = n - 2$$

$$= 7 - 2$$

$$= \underline{5}$$

④ find critical value at p=0.05

$r = 0.878$

n	$\alpha = 0.05$	$\alpha = 0.01$
4	0.950	0.990
5	0.878	0.959
6	0.811	0.917
7	0.754	0.875
8	0.707	0.834
9	0.666	0.798
10	0.632	0.765
11	0.602	0.735
12	0.576	0.708
13	0.553	0.684
14	0.532	0.661

if r value calculated is HIGHER than the value in the table

→ the two variables are correlated

→ Reject null hypothesis

difference is not due to random error/chance

if r value calculated is LOWER than the value in the table

→ the two variables are not correlated

→ Accept null hypothesis

difference is due to random error/chance

calculated
r value

table
r value

$$0.992 > 0.878$$

greater

∴ REJECT null hypothesis

Spearman's rank correlation coefficient:

the data should be ordinal, Not normally distributed, scatter graph shows a correlation but not linear & atleast 5 paired data

$$r_s = 1 - \left(\frac{6 \times \sum D^2}{n^3 - n} \right)$$

D = difference b/w each pair of RANKED measurements

n = # of pairs of observations

# of species R	Rank for species R	# of species S	Rank for species S	Rank difference (D)	D ²
38	7	24	6	1	1
2	1	5	2	-1	1
20	4.5	8	4	0.5	0.25
50	10	31	9	1	1
28	6	27	7	-1	1
8	2	4	1	1	1
42	8	36	10	-2	2
13	3	6	3	0	0
20	4.5	11	5	-0.5	0.25
43	9	30	8	1	1

Ranking
 → start by ranking the smallest value at 1 (small to big)
 → if there are two identical values, they share same rank (e.g.)

$$\sum D^2 = 8.5$$

① construct a null hypothesis

there is no significant correlation b/w the variables

② calculate the r_s value

$$r_s = 1 - \left(\frac{6 \times \sum D^2}{n^3 - n} \right)$$

$$r_s = 1 - \left(\frac{6 \times 8.5}{10^3 - 10} \right) \rightarrow r_s = 0.948$$

③ calculate the degrees of freedom

$$\begin{aligned} D.O.F. &= n - 2 \\ &= 10 - 2 \\ &= 8 \end{aligned}$$

sometimes you can just use the value of n if thats what the table says

④ find critical value at $p=0.05$

if r_s value calculated is HIGHER than the value in the table	if r_s value calculated is LOWER than the value in the table
→ the two variables are correlated	→ the two variables are not correlated
→ Reject null hypothesis	→ Accept null hypothesis
difference is not due to random error/chance	difference is due to random error/chance

calculated r_s value
 table r_s value
 $0.948 > 0.648$
 greater
 ∴ REJECT null hypothesis

not D.O.F!

	r_s	
	$P = 0.05$	$P = 0.01$
5	1.000	—
6	0.886	1.000
7	0.786	0.929
8	0.738	0.881
9	0.700	0.833
10	0.648	0.794
11	0.618	0.755
12	0.587	0.727
13	0.560	0.703
14	0.538	0.679
15	0.521	0.654
16	0.503	0.635
17	0.488	0.618